NDEJJE FARMERS SACCO

Loan Management Information System

TESTING EVIDENCE

Item	Details
Course unit	Object Oriented C++ (BSE 2102)
Programme	Bachelor of Software Engineering
Lecturer	Mr. Mwanje Abiyasaali
Group	Group 6
Members	MUGUME JULIUS (25/1/370/D/1497) KASIRYE CHRISTOPHER (25/1/370/D/2175)

1. How the testing was done.

The program was compiled in Visual Studio on Windows and run by the group. Each main feature was used the way a SACCO officer or member would use it and a screenshot was taken of the result. Under each screenshot we explain what was entered, what the system did and which requirement it proves.

2. Summary of the screenshots

No.	Test case	Expected result	Actual result	Status
1	Staff login	Admin logged in, password hidden	Logged in; password shown as ********	Pass
2	Add member record	Member registered	Member registered as NFS0002	Pass
3	View all members	Members listed with totals	Members listed with totals	Pass
4	Search record (valid ID NFS0002)	Record displayed	Record displayed	Pass
5	Search record (invalid ID NFS0009)	Error message displayed	Error message displayed	Pass
6	Member self-service login	Read-only member menu	Menu for Ssali Moses shown	Pass
7	Member account summary	Own details and limit shown	Details shown; limit UGX 600,000	Pass
8	Eligibility rules check	Each rule shown PASS/FAIL	6 PASS, 1 FAIL (membership period)	Pass
9	Input validation (negative amount)	Error handled/re-prompted	Error handled/re-prompted	Pass
10	Loan application, not eligible	Application refused	"Application NOT registered"	Pass
11	Approve screen, nothing pending	Clear message, no crash	"No loans found in this category"	Pass
12	Loan application, eligible	Saved as PENDING with cost shown	LN0001 PENDING; installment UGX 87,763	Pass
13	Approve loan	PENDING to APPROVED	"Loan LN0001 APPROVED"	Pass
14	Disburse approved loan	Cost summary and confirmation	Shown; total to repay UGX 526,576	Pass
15	Delete record	Record deleted	"Member NFS0003 has been deleted"	Pass

Screenshot 1: Staff login

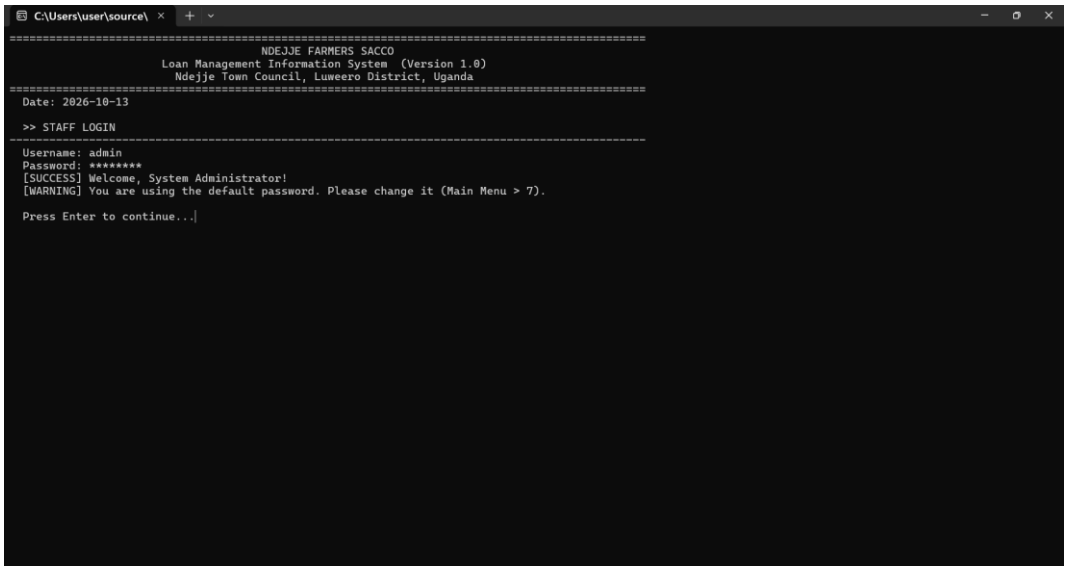


Figure 1: Staff login

What happened: The Administrator logs in with username admin. The password is hidden as ******** while typing and the system warns that the default password should be changed.

What it proves: Security: only staff with a valid username and password can use the system.

Result: **Pass**

Screenshot 2: Add member record

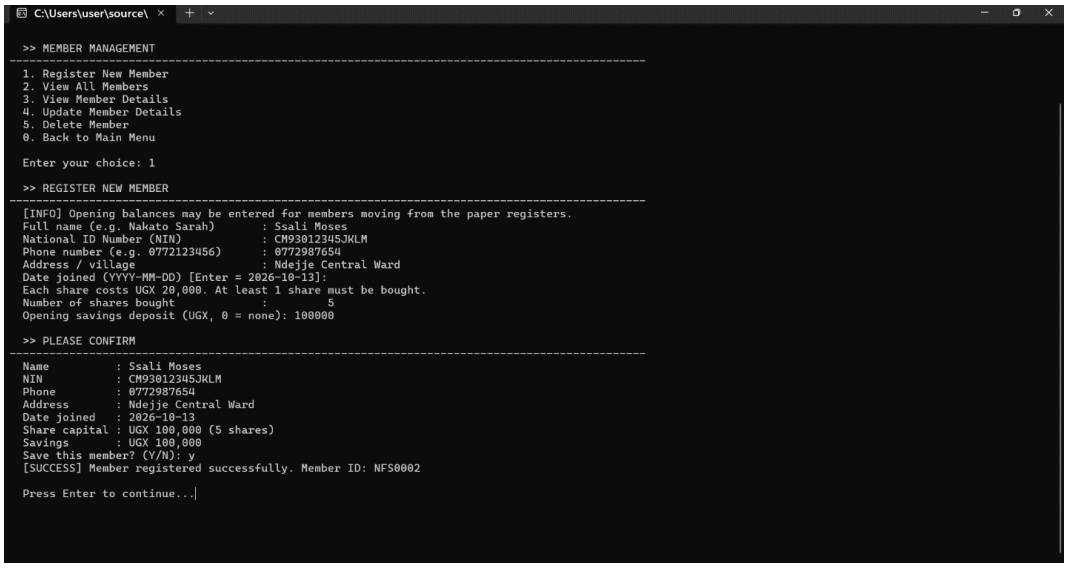


Figure 2: Add member record

What happened: A new farmer, Ssali Moses is registered with NIN, phone, address, 5 shares (UGX 100,000) and UGX 100,000 savings. The system shows a summary, asks for confirmation (Y) then gives the automatic Member ID NFS0002.

What it proves: Create: a member record is added and saved to the file ndejje_members.txt.

Result: **Pass**

Screenshot 3: View all members

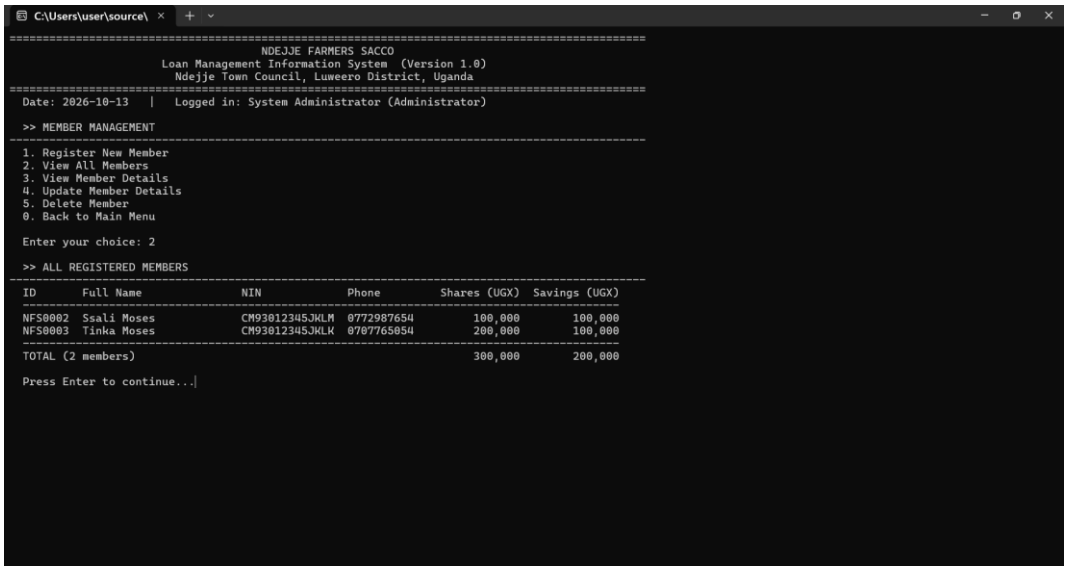


Figure 3: View all members

What happened: The member register lists every member with ID, name, NIN, phone, share capital and savings, with totals at the bottom.

What it proves: Read: records are loaded from the file and totals are calculated in a loop.

Result: **Pass**

Screenshot 4: Search record with a valid Member ID

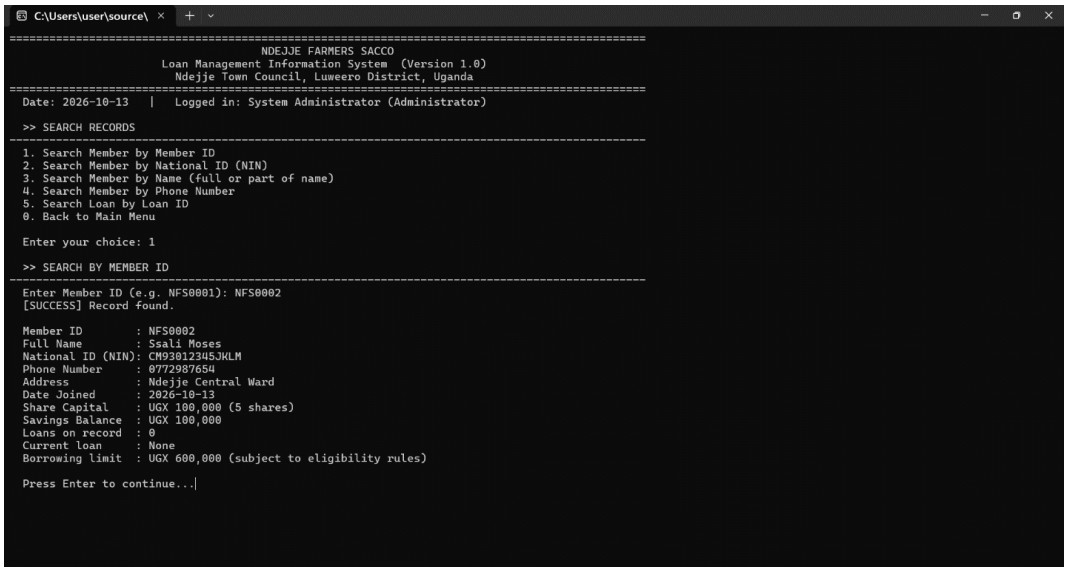


Figure 4: Search record with a valid Member ID

What happened: In Search Records, the Member ID NFS0002 was entered. The system shows 'Record found' and displays Ssali Moses' full record and borrowing limit.

What it proves: Search: a valid ID returns the correct record.

Result: **Pass**

Screenshot 5: Search record with an invalid Member ID

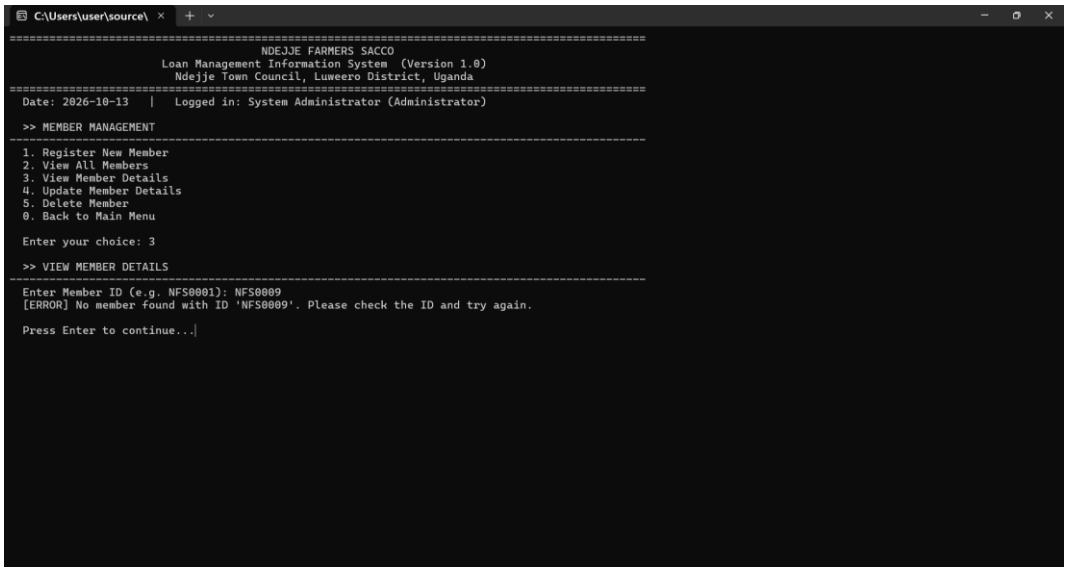


Figure 5: Search record with an invalid Member ID

What happened: The ID NFS0009 does not exist. The system shows a clear error message instead of crashing, then waits for Enter.

What it proves: Search + error handling: a wrong ID is handled safely.

Result: **Pass**

Screenshot 6: Member self-service login

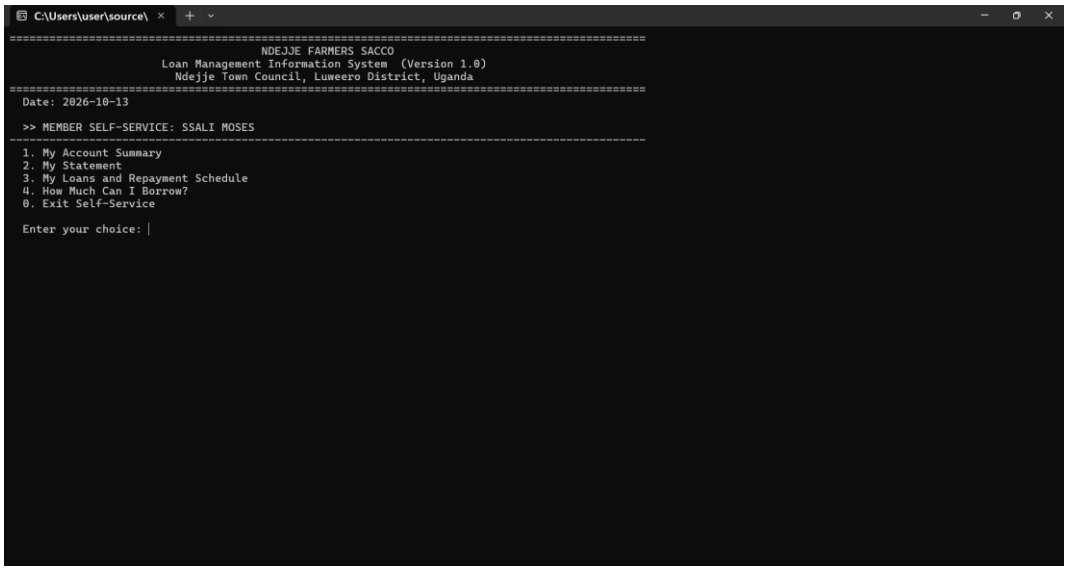


Figure 6: Member self-service login

What happened: Ssali Moses opened his own account using his Member ID and NIN. He gets a read only menu: summary, statement, loans and borrowing limit.

What it proves: Security: a member can view but never change their own data.

Result: **Pass**

Screenshot 7: Member views own account summary

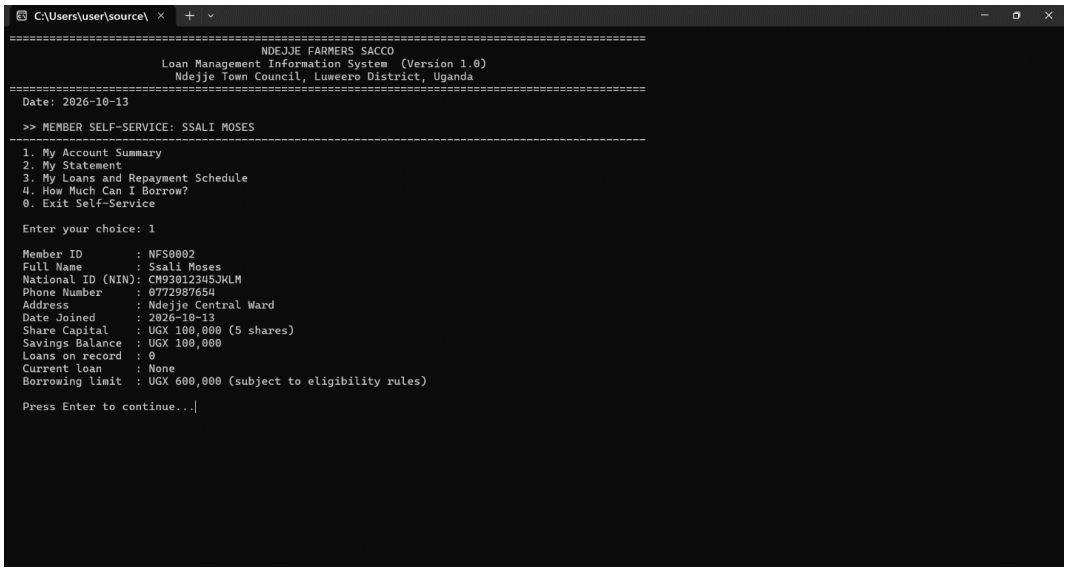


Figure 7: Member views own account summary

What happened: Option 1 shows the member's details, balances, loans and borrowing limit (3 x (savings + shares) = UGX 600,000).

What it proves: Calculation: the borrowing limit is worked out automatically.

Result: **Pass**

Screenshot 8: Borrowing limit and eligibility rules

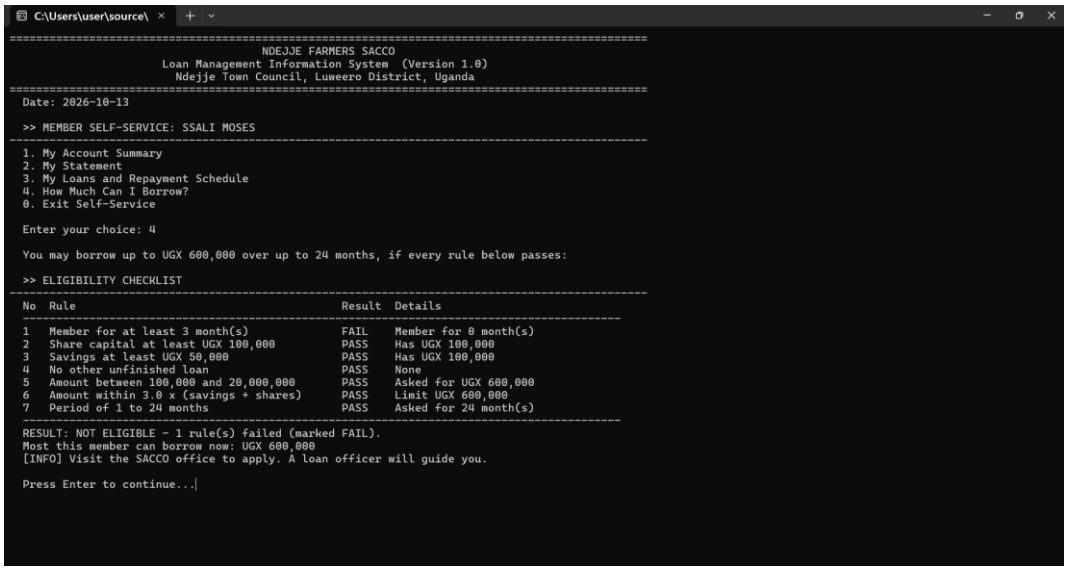


Figure 8: Borrowing limit and eligibility rules

What happened: The system checks 7 SACCO rules. Ssali Moses joined today so rule 1 (member for at least 3 months) shows FAIL; the other 6 rules PASS.

What it proves: Decision-making: each rule is an if-condition; one FAIL makes the result NOT ELIGIBLE.

Result: **Pass**

Screenshot 9: Input validation: negative loan amount

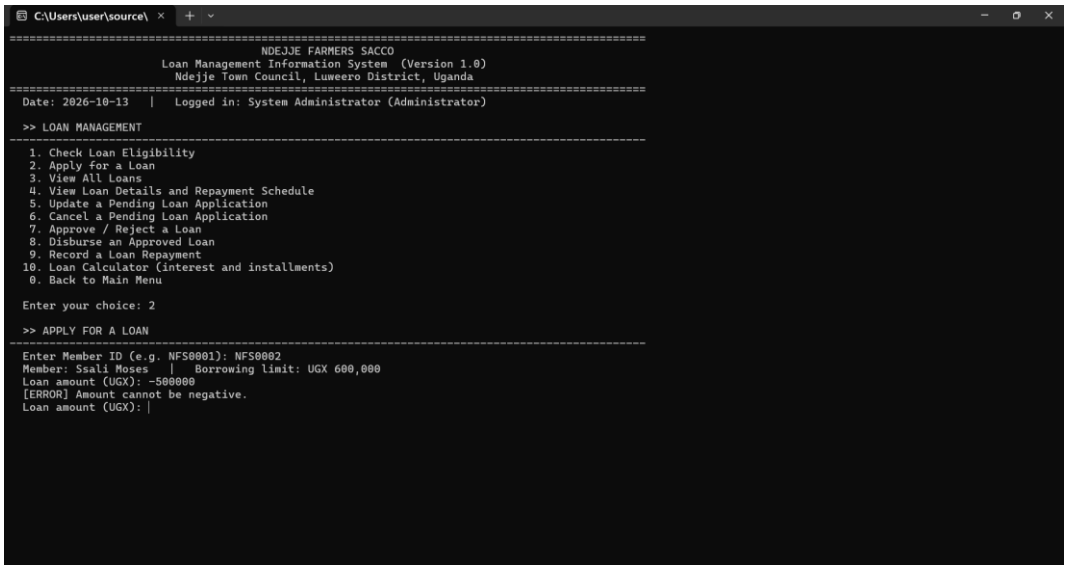


Figure 9: Input validation: negative loan amount

What happened: At 'Loan amount' the user typed -500000. The system rejected it with 'Amount cannot be negative' and asked again.

What it proves: Validation: negative money is never accepted and the program does not crash.

Result: **Pass**

Screenshot 10: Loan application refused (member not eligible)

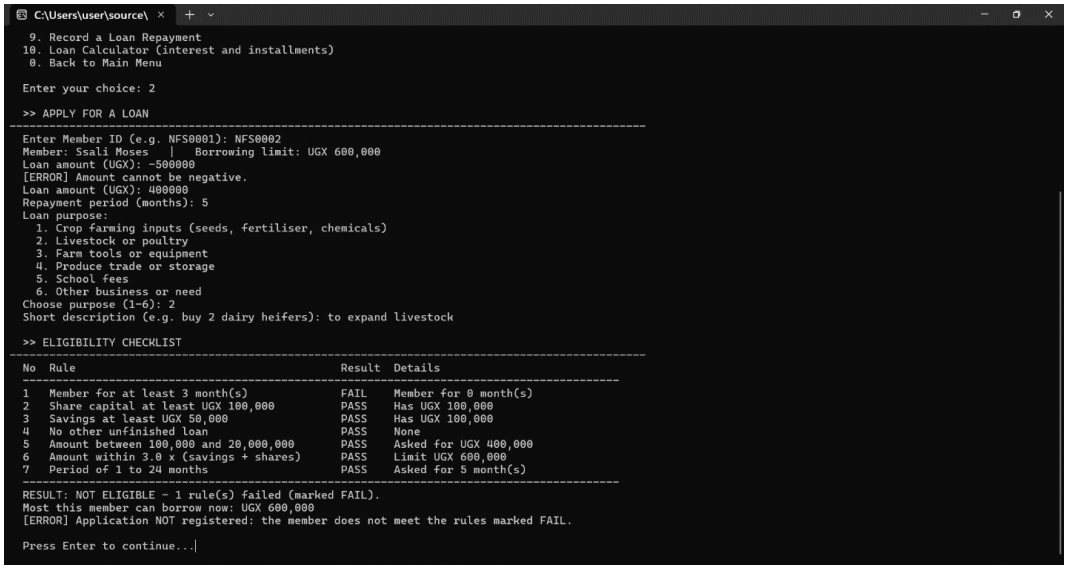


Figure 10: Loan application refused (member not eligible)

What happened: With a valid amount (UGX 400,000, 5 months, livestock), rule 1 still fails so the application is NOT registered.

What it proves: Business rules: loans are only recorded for members who meet every rule.

Result: **Pass**

Screenshot 11: Approve screen with no pending loans

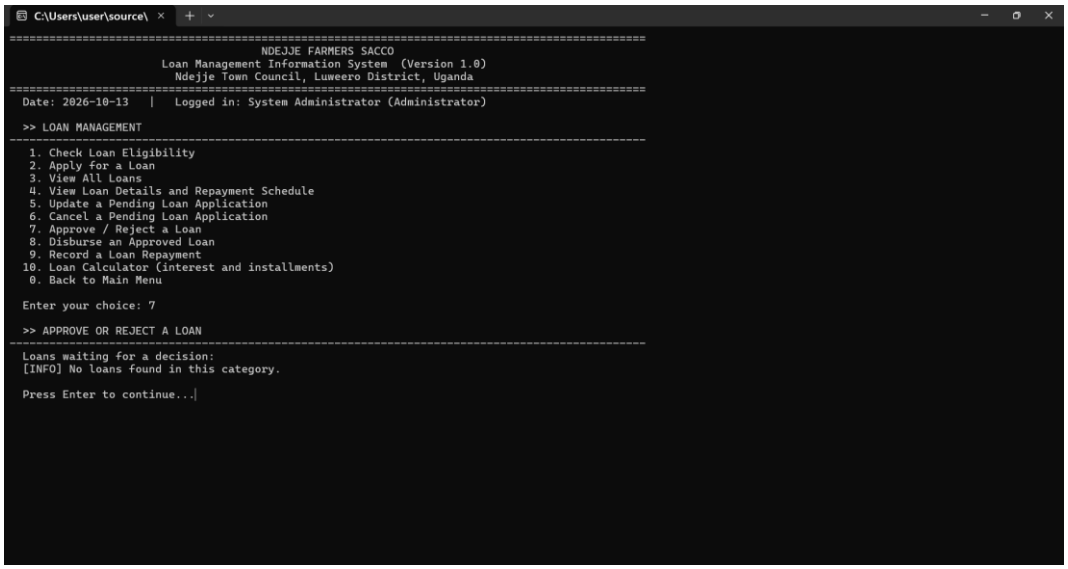


Figure 11: Approve screen with no pending loans

What happened: Because the application above was refused, there is no PENDING loan to approve. The system says so instead of failing.

What it proves: Error handling: an empty list is handled with a clear message.

Result: **Pass**

Screenshot 12: Loan application accepted (PENDING)

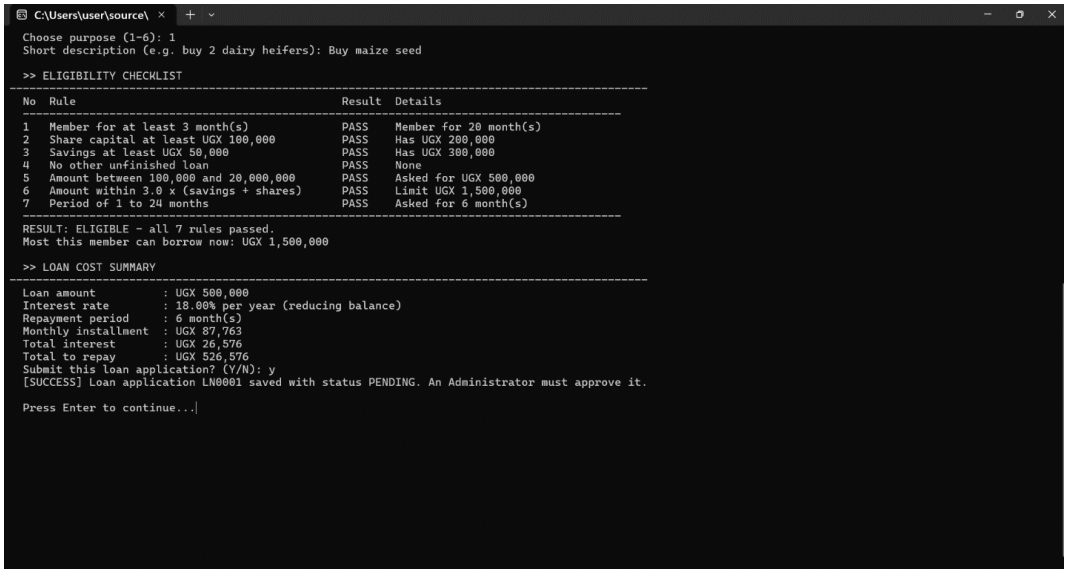


Figure 12: Loan application accepted (PENDING)

What happened: Nakato Sarah (NFS0004) was registered from the paper records with a join date 20 months ago, 10 shares and UGX 300,000 savings. She applied for UGX 500,000 over 6 months to buy maize seed. All 7 rules PASS, the system shows the cost (installment UGX 87,763, interest UGX 26,576) and saves loan LN0001 as PENDING.

What it proves: Loan application and automatic interest calculation (reducing balance).

Result: **Pass**

Screenshot 13: Approve the loan

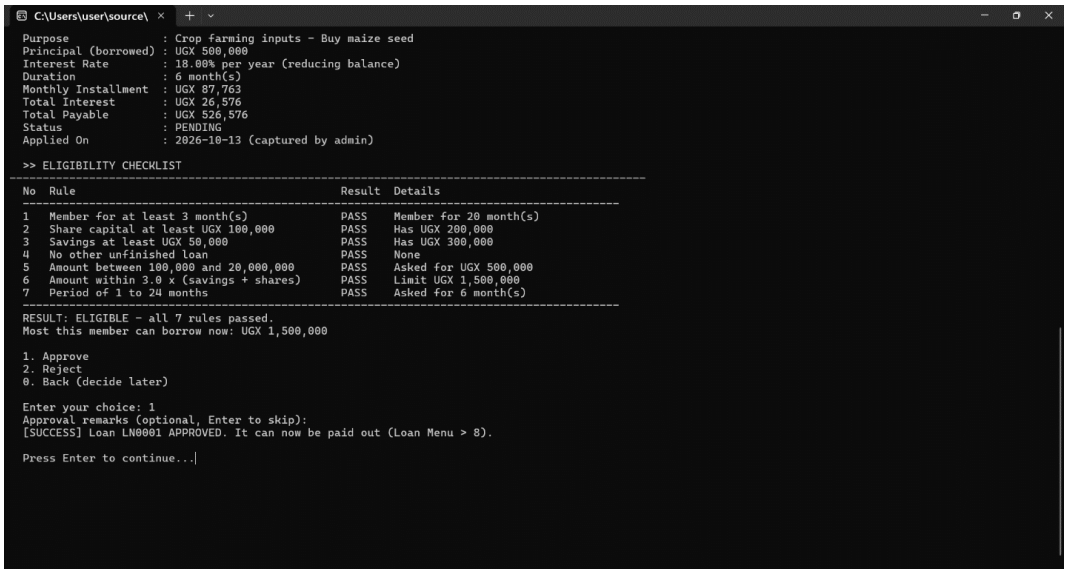


Figure 13: Approve the loan

What happened: The Administrator opens LN0001. The eligibility checklist is checked again, all 7 rules pass, and the loan is approved.

What it proves: Loan workflow: status changes from PENDING to APPROVED. Only an Administrator can approve.

Result: **Pass**

Screenshot 14: Disburse (pay out) the approved loan

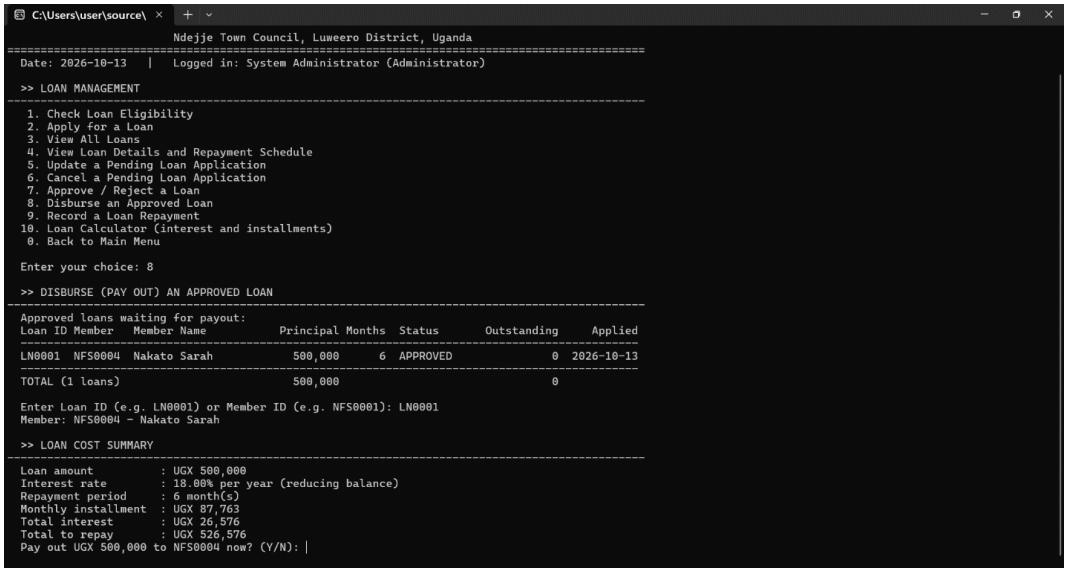


Figure 14: Disburse (pay out) the approved loan

What happened: LN0001 appears in the list of approved loans waiting for payout. The system shows the cost summary (UGX 500,000 borrowed, UGX 526,576 to repay) and asks for confirmation before paying out.

What it proves: Loan workflow: only APPROVED loans can be paid out; the payment amounts are calculated automatically.

Result: **Pass**

Screenshot 15: Delete record

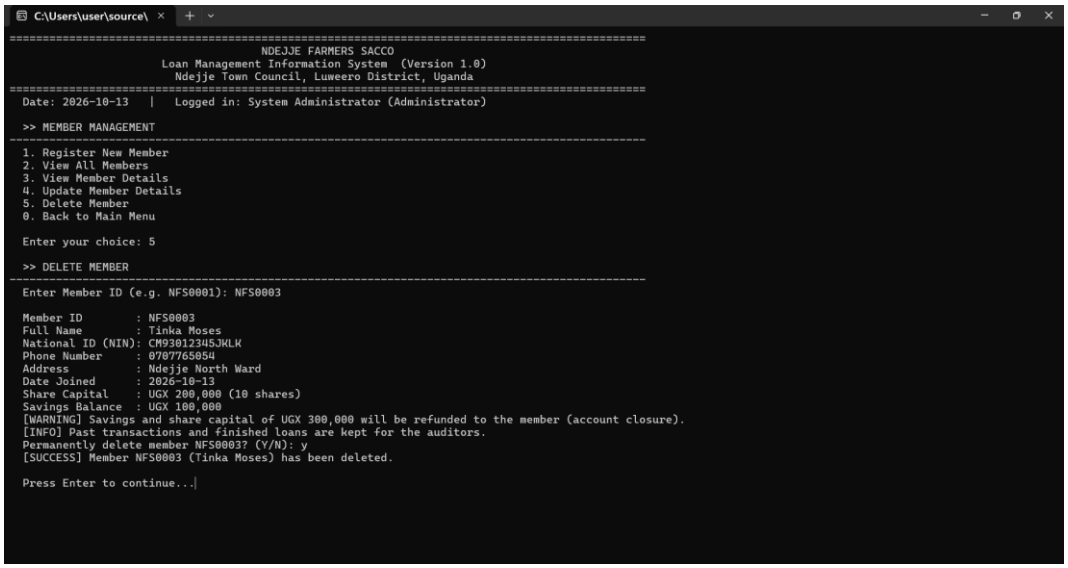


Figure 15: Delete record

What happened: Member NFS0003 (Tinka Moses), who has no loan, is deleted. The system warns that UGX 300,000 (savings + shares) will be refunded and asks for confirmation before deleting.

What it proves: Delete: the record is removed from the file; the refund is kept in the ledger for auditors.

Result: **Pass**